

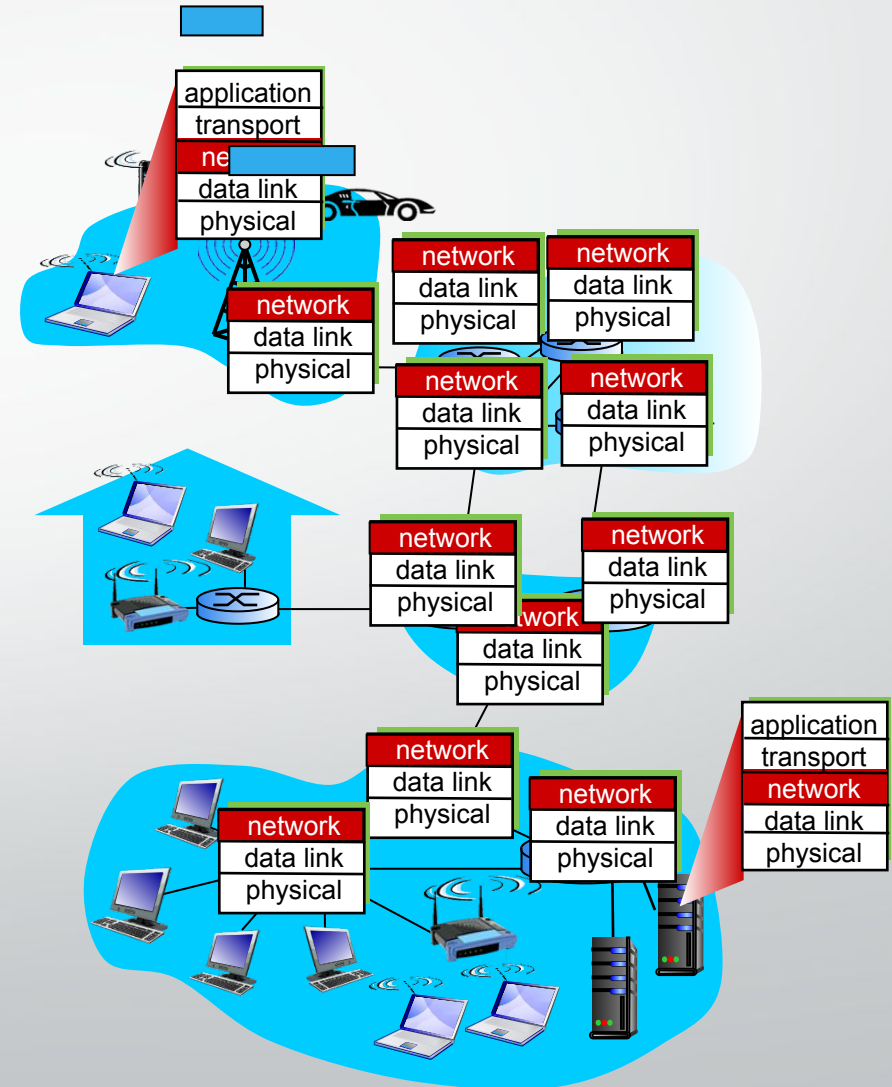
Objectives

- Short overview of the Network Layer
- Packet Switching: Virtual Circuits & Datagram Networks
- IPv4 Packet Format
- IP Fragmentation & Reassembly
- ICMP
 - Ping
 - Traceroute

The Network Layer

- Encapsulates data into **packets** on the sending side.
- Network Layer protocols operates on **hosts** and **routers**.
- **Routers** inspect IP header fields for forwarding.

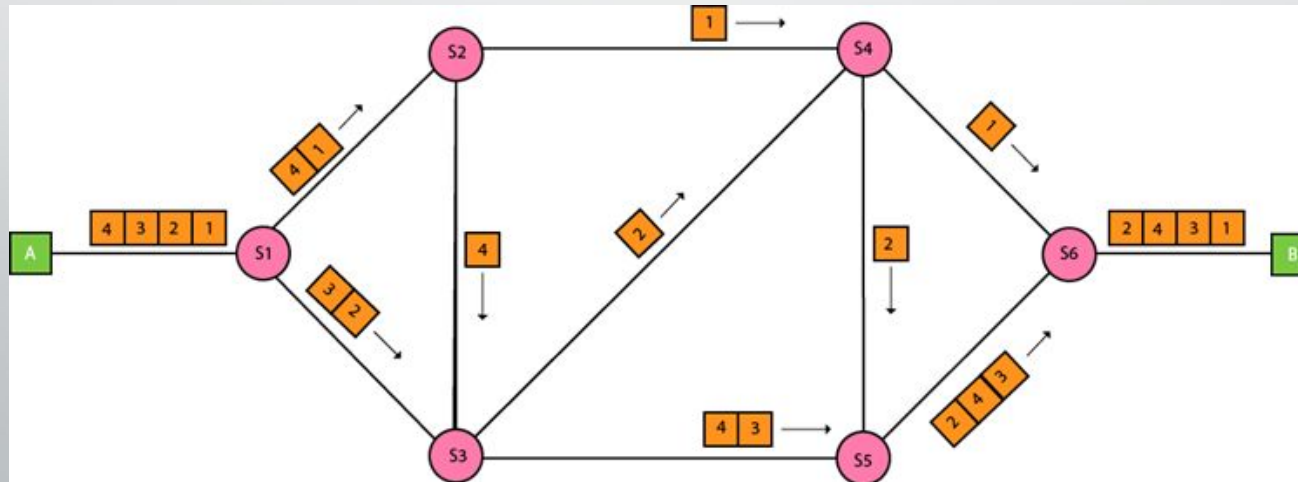
Delivers segments to the **transport layer** on the



Packet Switching

Packet Switching

- Packet Switching is a method of transferring data across a network by breaking it into smaller packets.
- Two type of networks based on packet switching
 - **Datagram Networks**
 - **Virtual Circuit Networks**



Datagram networks

- **No Call Setup:**

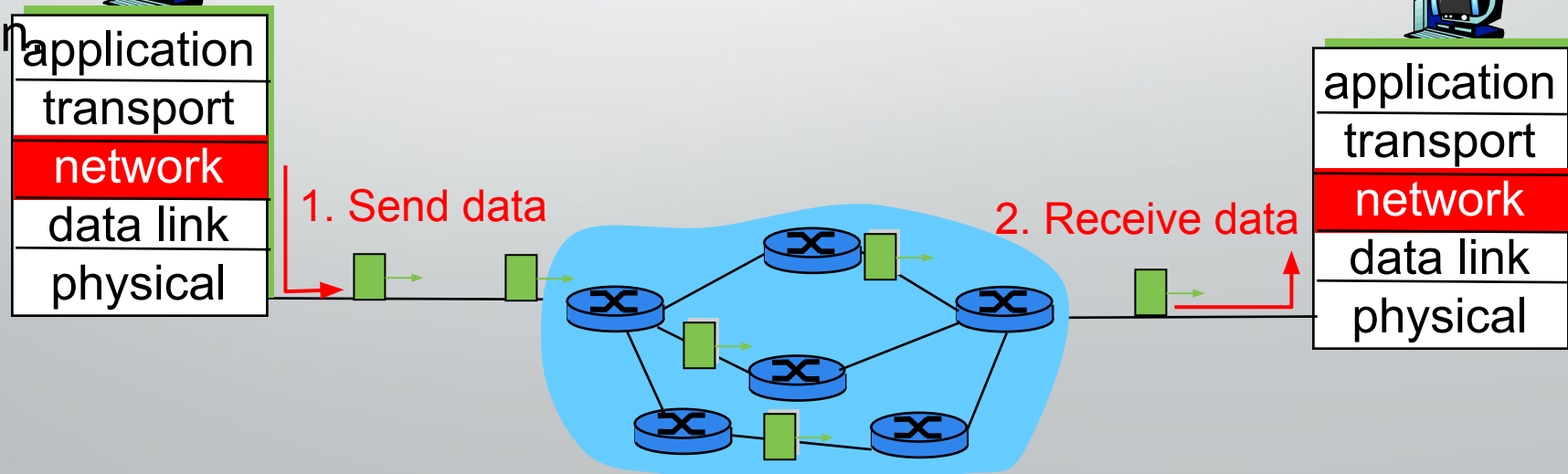
Devices can send data immediately without establishing a connection.

- **Stateless Routers:**

Routers forward packets independently based on their destination IP address.

- **Packet Forwarding:**

Packets from the same source may take different paths to reach the destination.



Virtual Circuits: Signaling Protocols

- **Call Setup:**

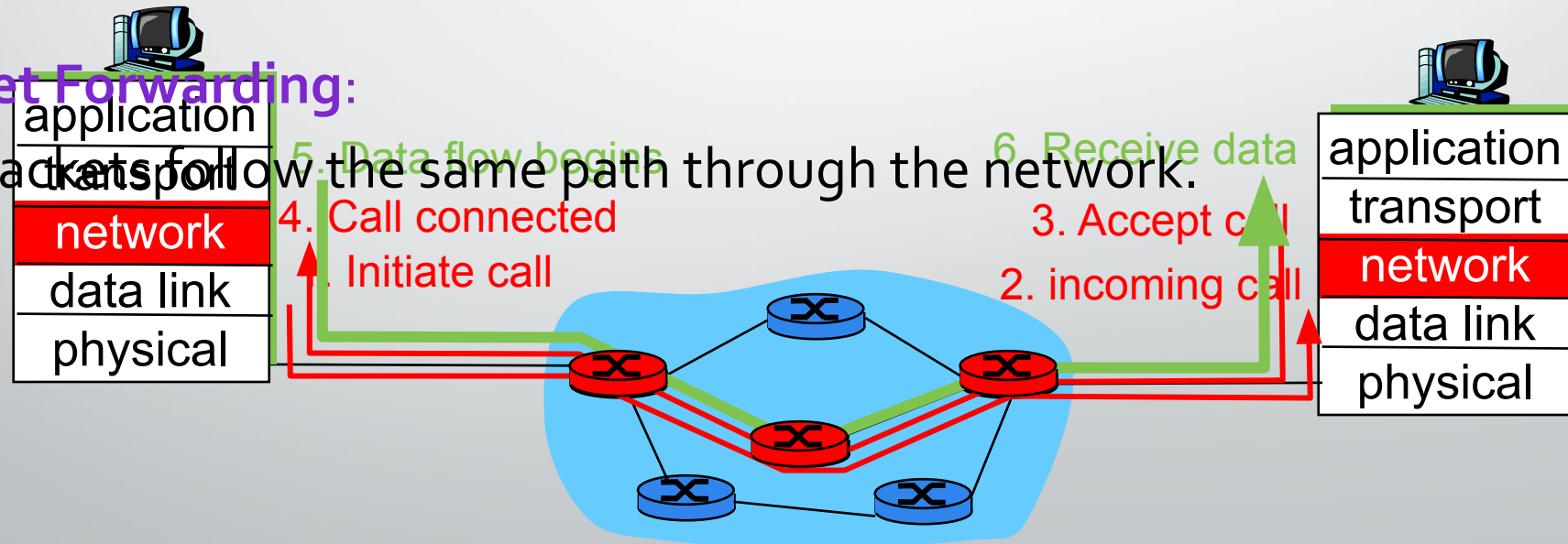
A connection (virtual circuit) is established between sender and receiver before data transfer.

- **Stateful Routers:**

Routers maintain information about active connections (virtual circuits).

- **Packet Forwarding:**

All packets follow the same path through the network.





Inspiring Excellence

Network Layer: IPv4 Functions

Lecture 9 | CSE421 – Computer Networks

Department of Computer Science and Engineering
School of Data & Science

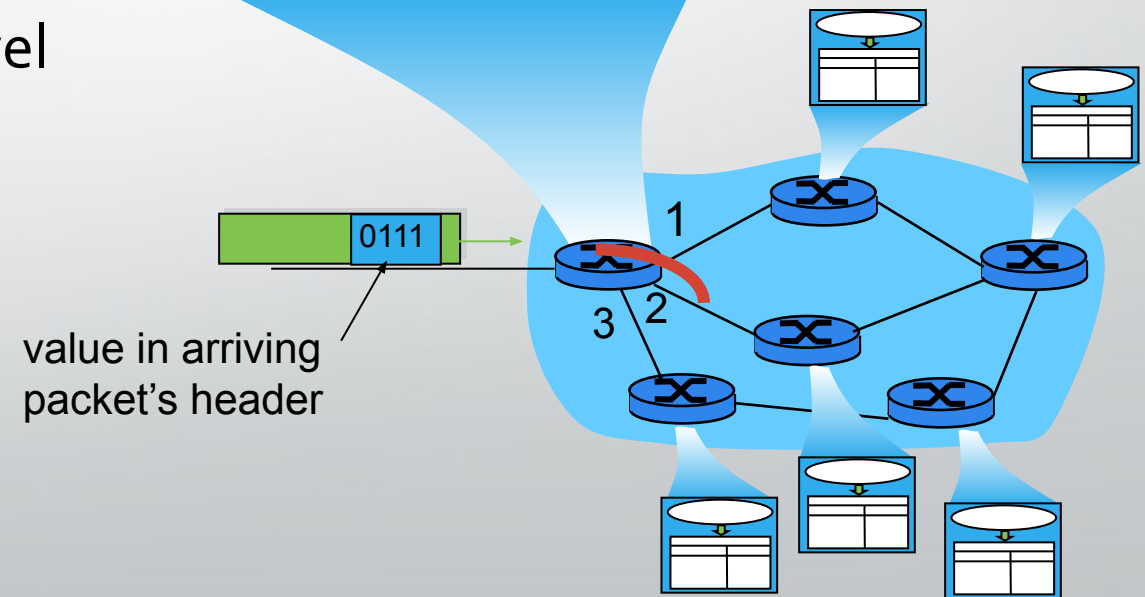
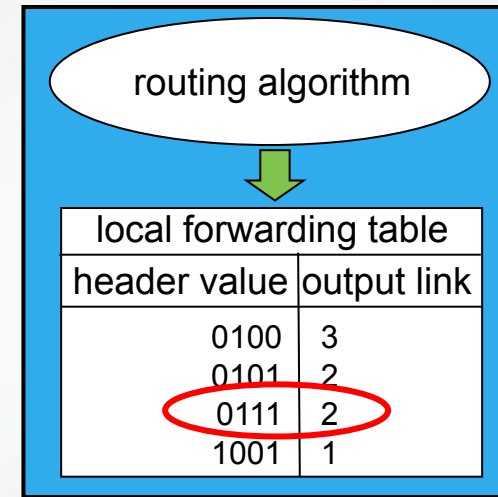
Functions of Network Layer

- **Routing:**

- Finds the best path from source → destination
- Done by routing algorithms
- Analogy: planning a trip before you travel

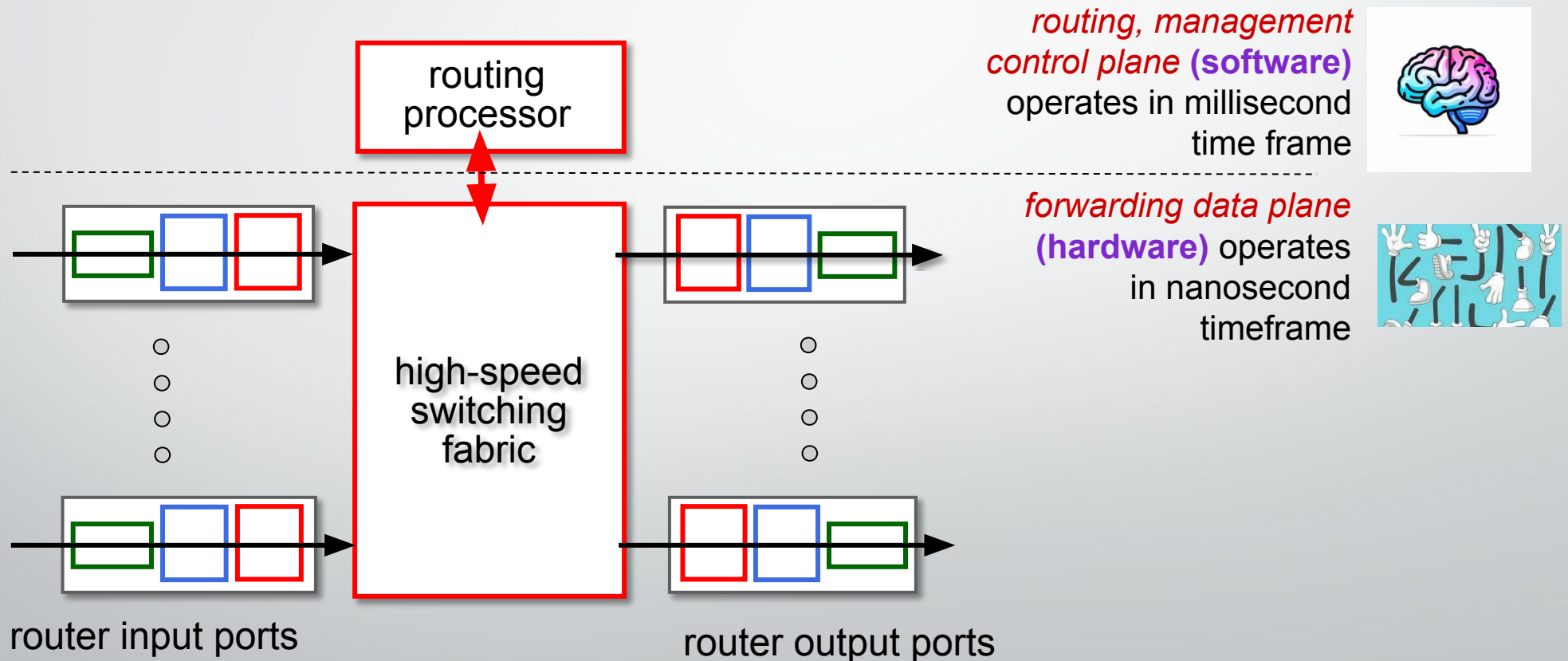
- **Forwarding:**

- move packets from router's input to appropriate router output
- Analogy: process of getting through a single interchange



Router architecture overview

high-level view of generic router architecture:



Control Plane – the brain

- Control plane builds the **routing table** using routing protocols.
- It creates the **forwarding table FIB** (Forwarding Information Base), a simplified version of routing table.
- The FIB may be created using
 - simple destination-based rules
 - or more advanced generalized rules.

Control plane installs this FIB into every input port.

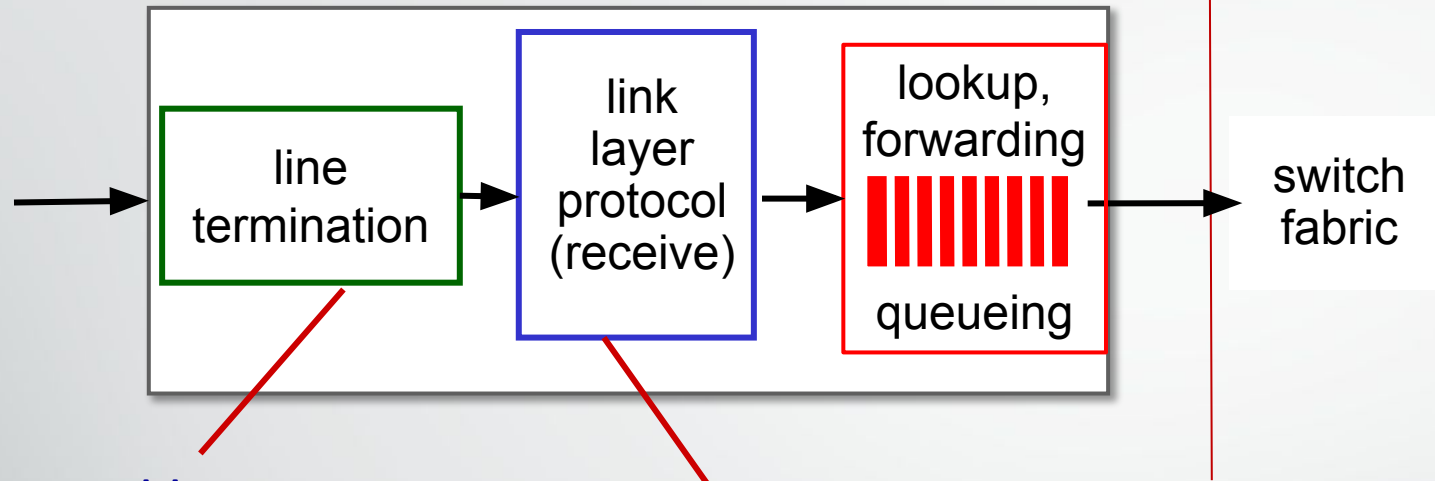
Decentralized Switching

- How input ports use the FIB?

Data Plane – the hands and legs!

- Forwarding rules can be:
 - **Destination-based forwarding :**
 - Uses **only destination IP** to choose output port
 - If destination = 200.20.20.0/24 → send to port so/o/o.
 - **Generalized forwarding :**
 - Admins can add **multiple header fields** (IP, protocol, port, etc.) Or special rules (QoS)
 - If video packet (UDP, port 5000) → high-priority queue.
- These rules are stored in the FIB created by the control

Input port functions



physical layer:

- Converts signal → digital bits

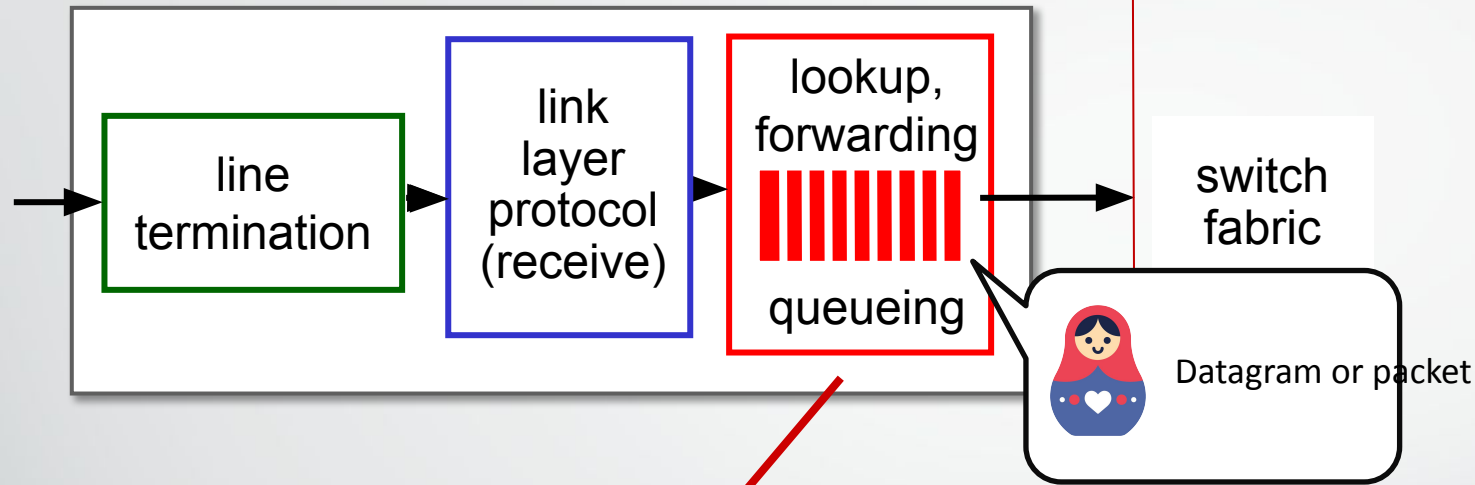
link layer:

- Removes data link header/trailer
- Checks for errors
- Extracts the IP packet



frame

Input port functions



Network layer:

- Input port checks packet header
- Matches against local FIB copy
- Decides output port (using destination-based or generalized rule)

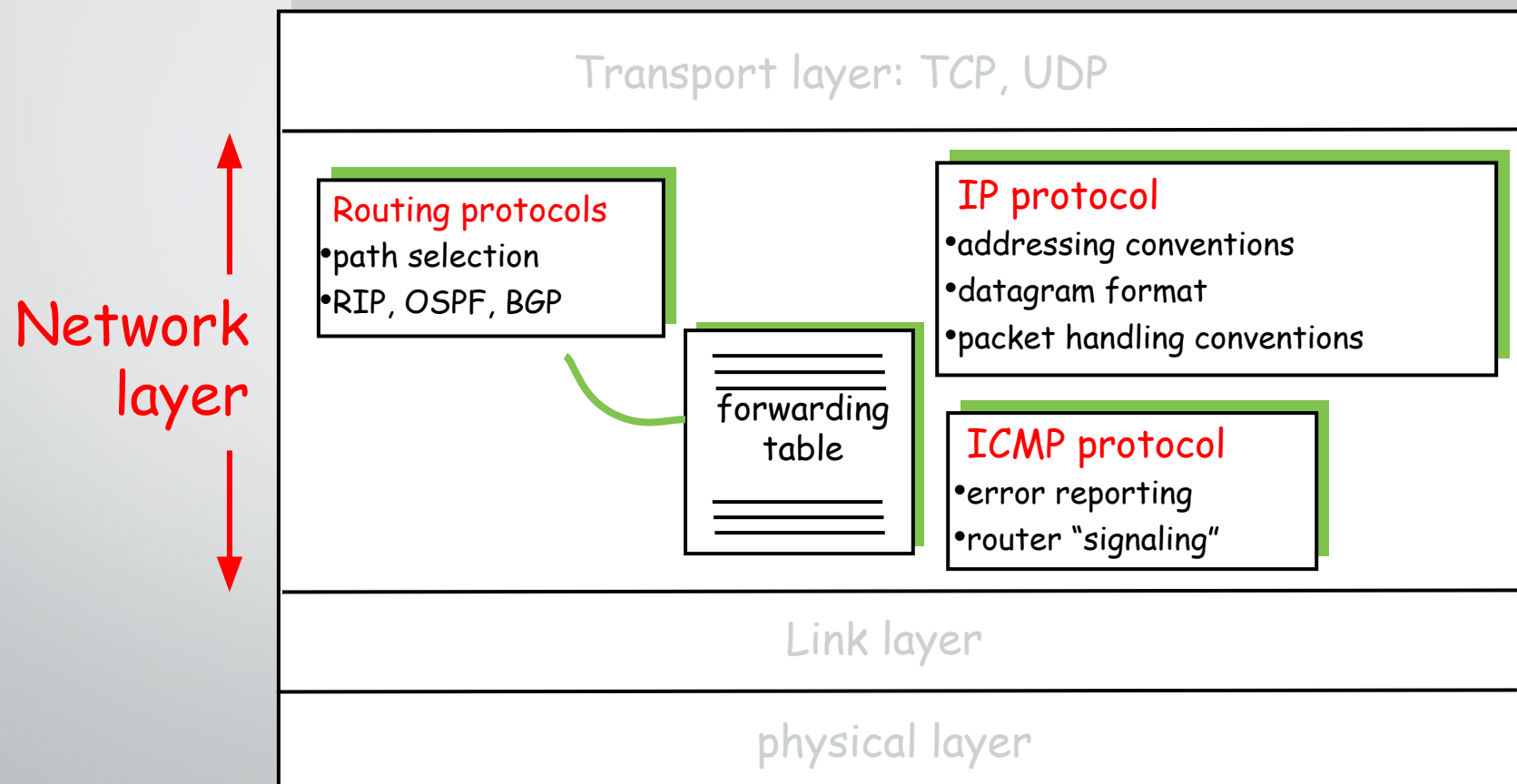
Input Port Queueing

- If packets arrive too fast → they wait here
- If queue is full → packet is dropped

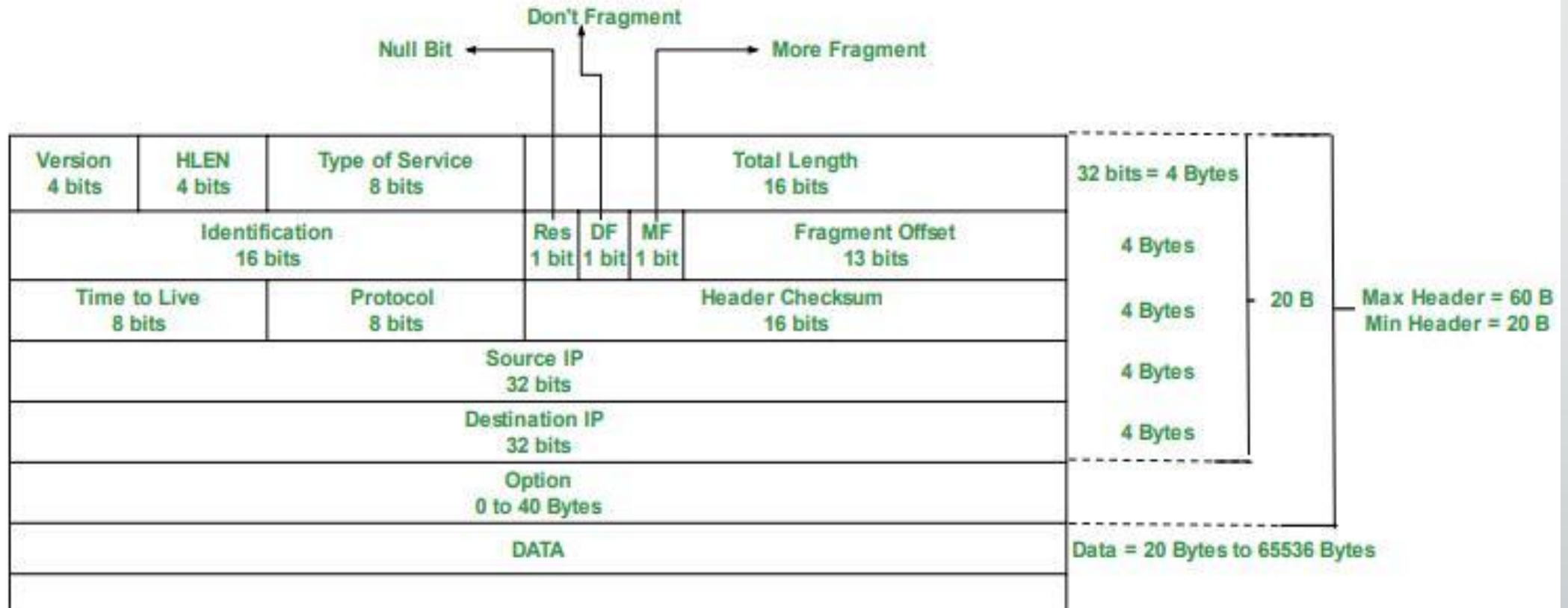
Internet Protocol IPv4

Internet Network Layer

- Host, router network layer functions:



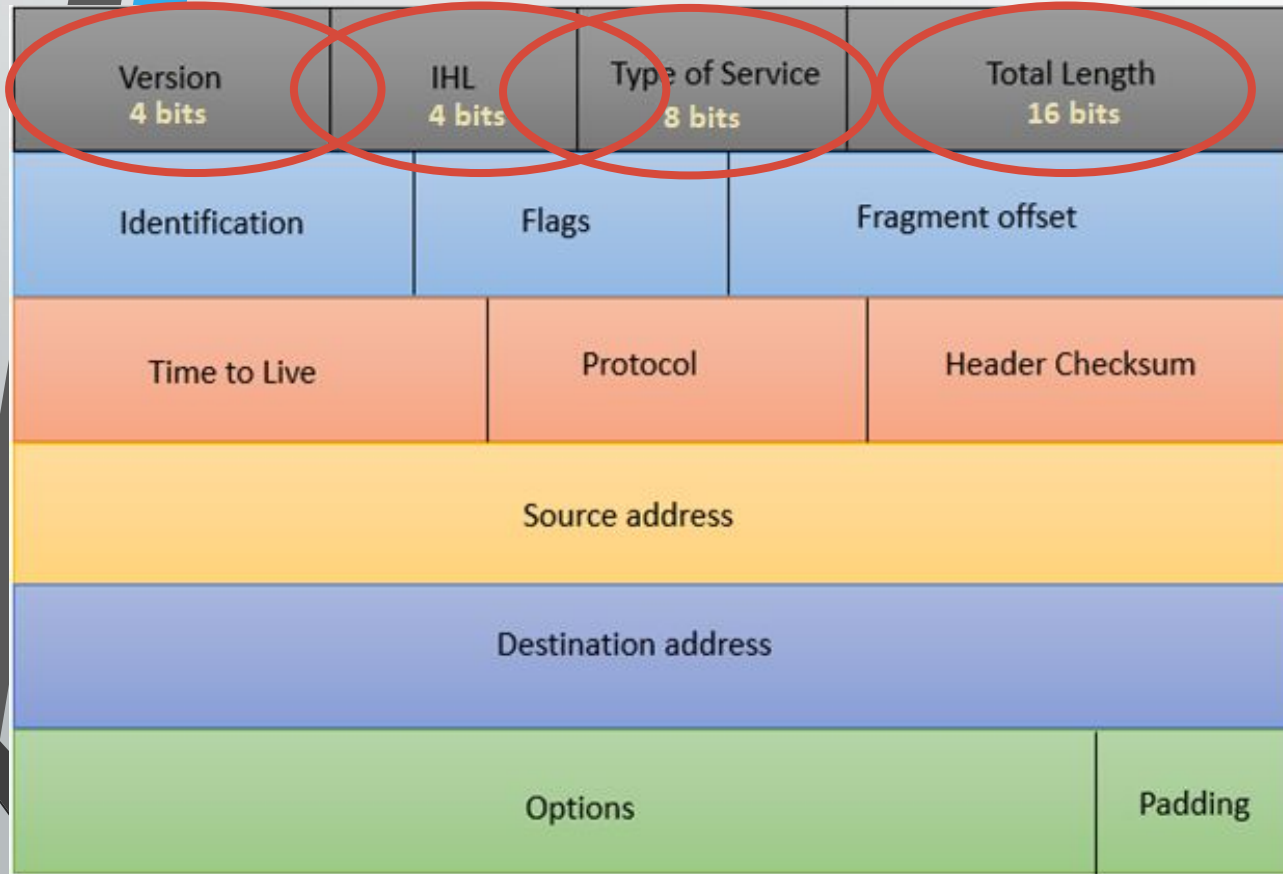
IPv4 Datagram Format



The size of an IP datagram:

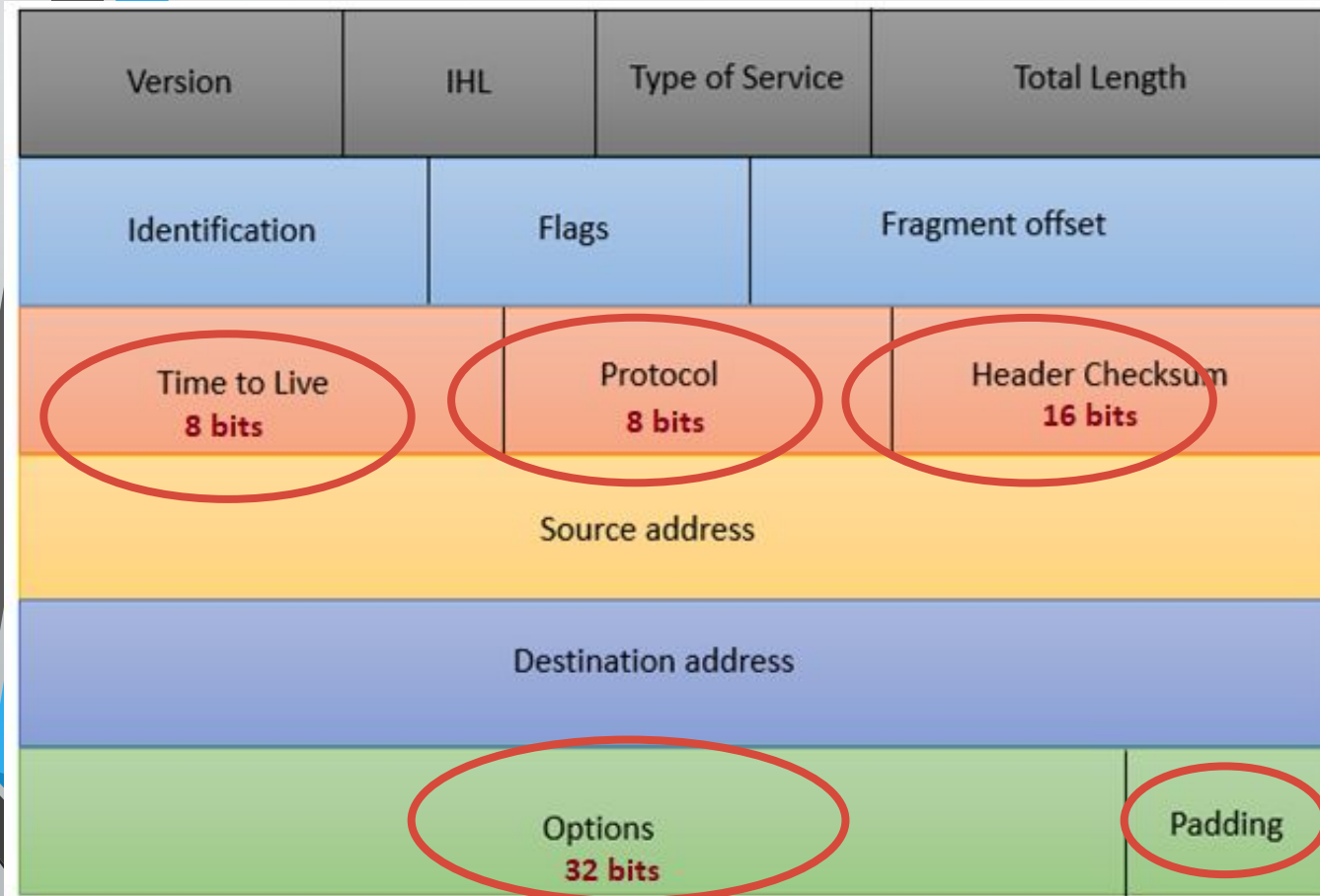
- The **minimum** size is **20 bytes** (if you have no data)
- The **maximum** size is **65,535 bytes**

IPv4 Datagram Header Format



- **Version:** value of which IP version is being used. For IPv4 the value will be 4 here.
- **Internet Header Length:** value of the header length, min 20 bytes, max 60 bytes. Shown in 4 byte word. **So min value 5, max 15.**
- **Type of Service:** for QoS (Quality of Service). To mark the packet to give special treatment or priority.
- **Total Length:** value of the entire size of the IP packet (header and data) in bytes.

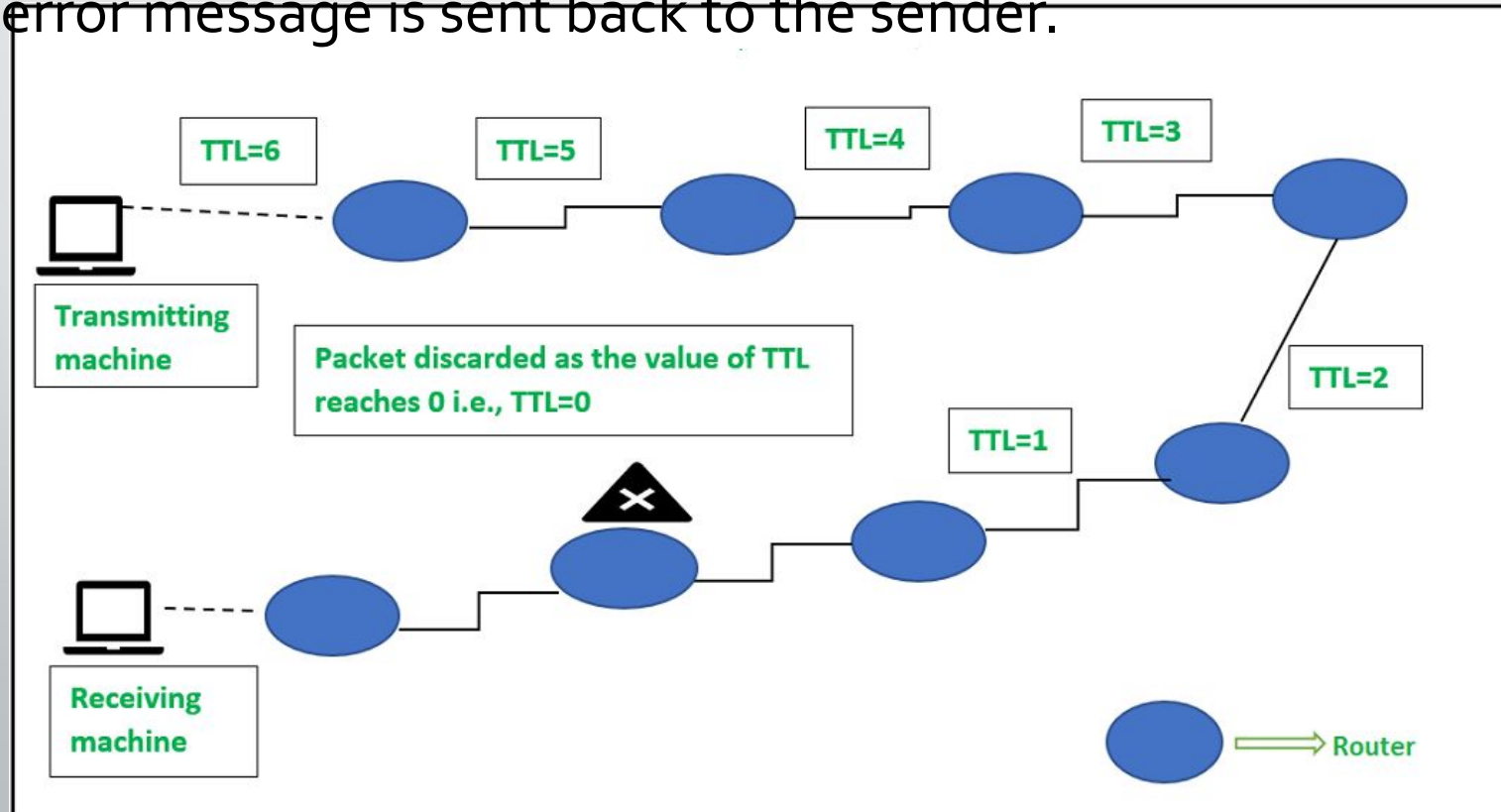
IPv4 Datagram Header Format



- **Time to Live:** maximum number of **hops** (routers) a packet can travel
- **Protocol:** value tells us which upper layer protocol is present, for example **TCP** has value **6** and **UDP** has value **17**.
- **Header Checksum:** to check if there are any errors in the header.
- **Options:** value of any extra information. Options are rarely used now.
- **Padding:**
Used only when Options are present. Ensures the header length becomes a multiple of 4 bytes

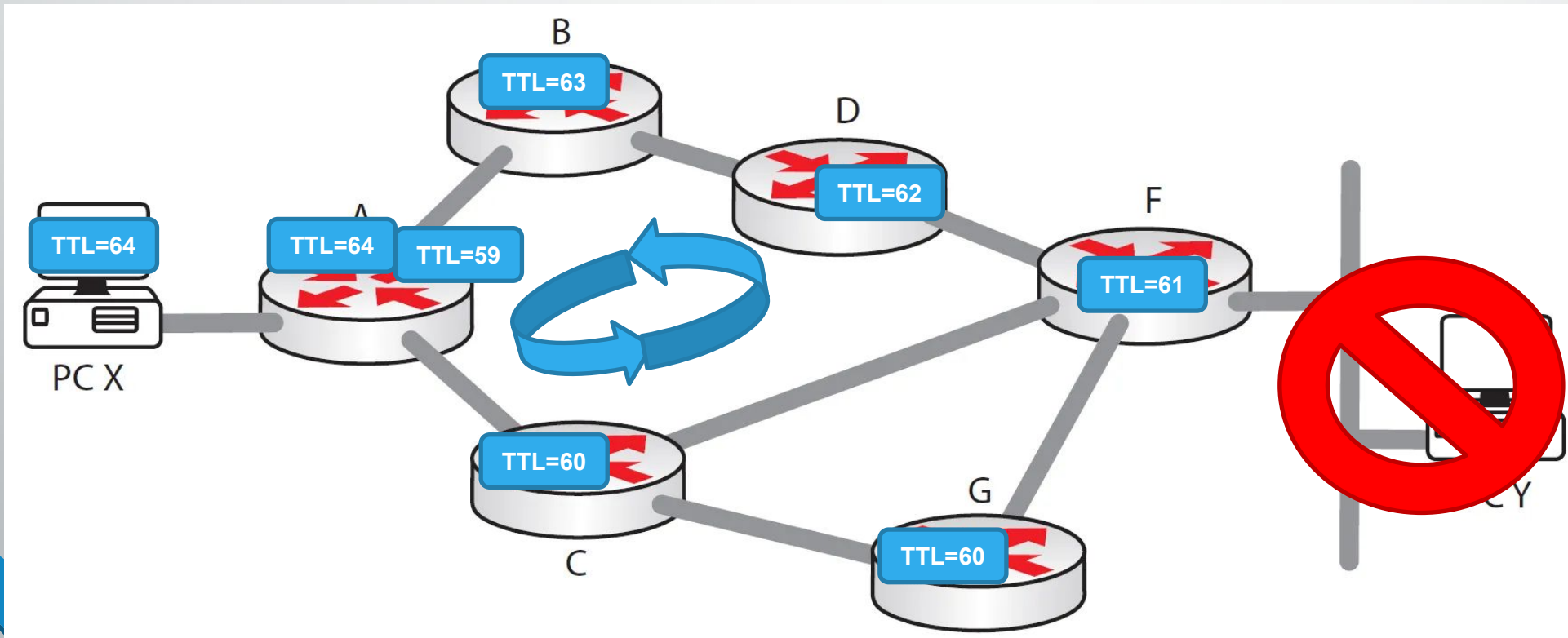
Time to Live - TTL

- Maximum number of **hops** (routers) a packet can traverse before being discarded.
- At each hop, the TTL is decreased by **1**.
- When the **TTL reaches 0**, the packet is dropped.
- And an error message is sent back to the sender.

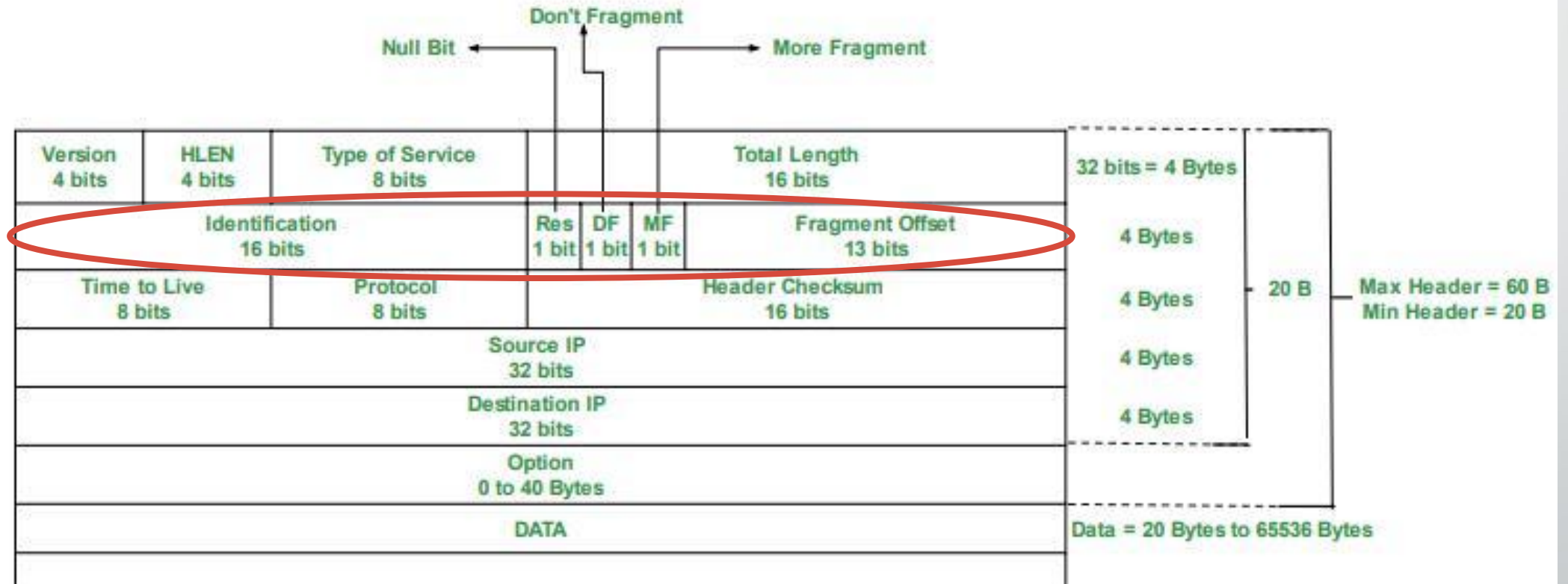


Time to Live - TTL

- **Not** just the "value of hops"
- It's a mechanism to prevent packets from **looping endlessly** in the network.
- Ensure finite packet lifetimes.

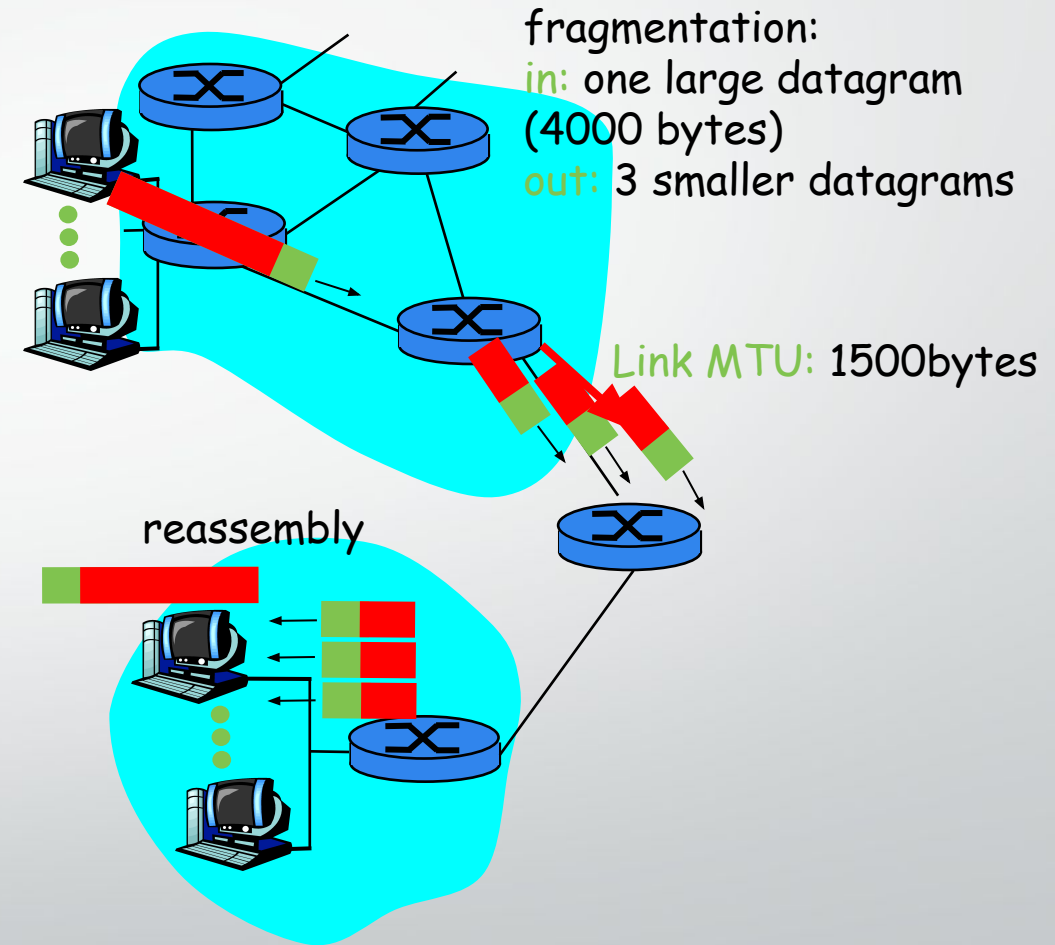


IPv4 Datagram Format



IP Fragmentation & Reassembly

- Network links have **MTU**
 - Maximum transmission unit or maximum transfer size
 - Different link types have different MTUs



IP Fragmentation & Reassembly

Original IP Datagram

Identifier	Total Length	DF May / Don't	MF Last / More	Fragment Offset
345	5140	0	0	0

Data 5140 => 20(H)+5120(D)

MTU=1500 => 20(H)+1480(D)

1st fragment : 5120-1480=3640

2nd fragment : 3640-1480=2160

3rd fragment : 2160-1480=680

Last/4th Segment : 680+20=700

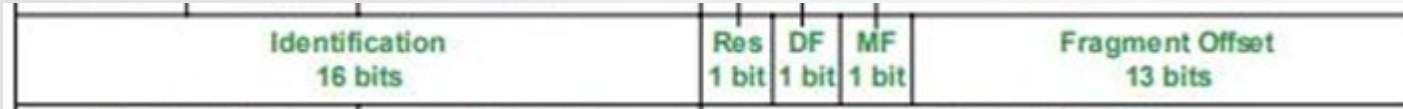
IP Fragments (Ethernet)

Identifier	Total Length	DF May / Don't	MF Last / More	Fragment Offset
345	1500	0	1	0
345	1500	0	1	185
345	1500	0	1	370
345	700	0	0	555

1st / start byte number

Data Bytes	Fragment Offset
0-1479	0/8=0
1480-2959	1480/8=185
2960-4439	2960/8=370
4440-5119	4440/8=555

IP Fragmentation & Reassembly



Extra Example

- Example:
 - 4000 Bytes of datagram
 - MTU = 1500 Bytes
- **DF – Don't Fragment Bit**
 - Value 0 or 1
- **Fragment Offset**
 - The value of the offset is measured in units of 8 bytes.

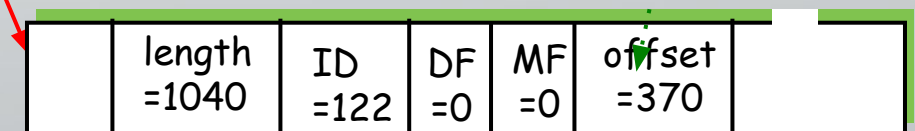
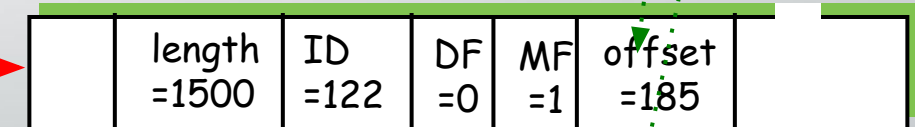
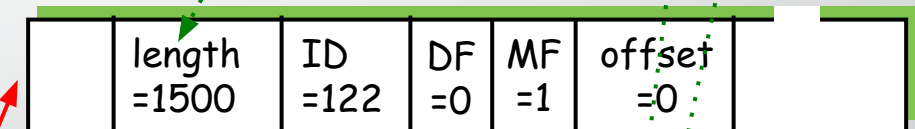


One large datagram becomes several smaller datagrams

1480 bytes in data field

offset = 1480/8

offset = 2960/8



ICMP

Internet Control Message Protocol

ICMP

- It helps devices send error messages and status updates.
- **Functions:**
 - Reporting errors in the network
 - Checking reachability (Is the host alive?)
 - Diagnosing delays and congestion
- **Key Points:**
 - ICMP does not carry actual user data
 - Mainly used by the **operating systems** for network management
 - **Example of ICMP in practice**
 - Ping
 - Traceroute



Ping

- Packet Internet Groper and is a network utility tool.

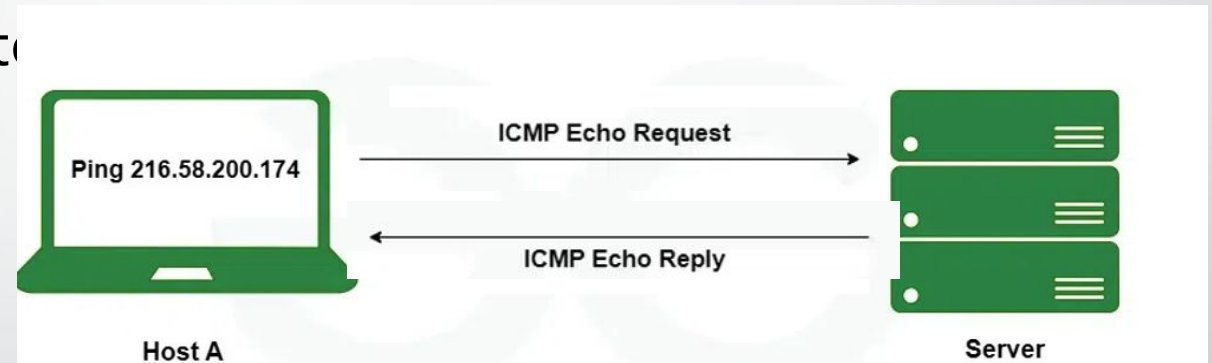
- **Purpose:**

- Checks if a device is reachable
- Measures the time it takes for data to

- **Mechanism:**

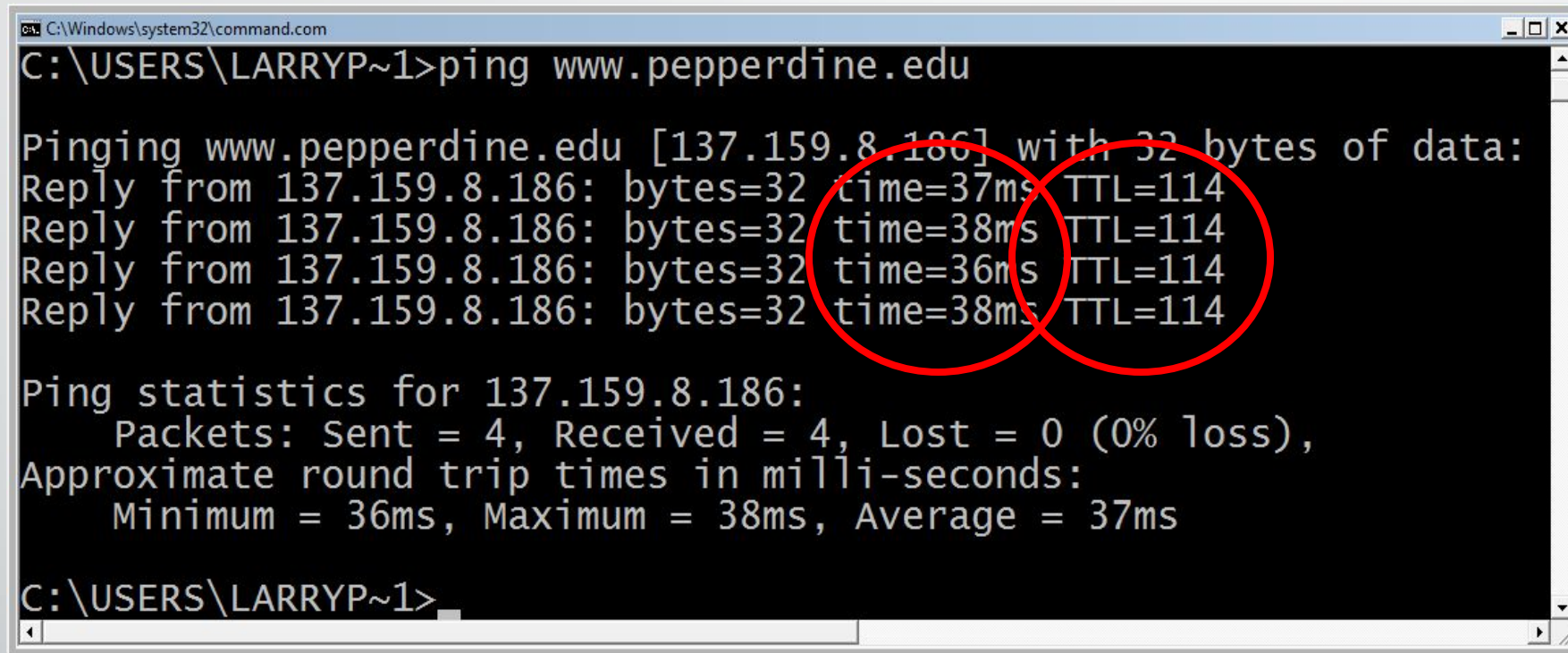
- Sends ICMP Echo Request packets
- Receives Echo Reply packets.

- **Results:**



- Commands:
ping 216.58.200.174

Ping



```
C:\Windows\system32\command.com
C:\USERS\LARRYP~1>ping www.pepperdine.edu

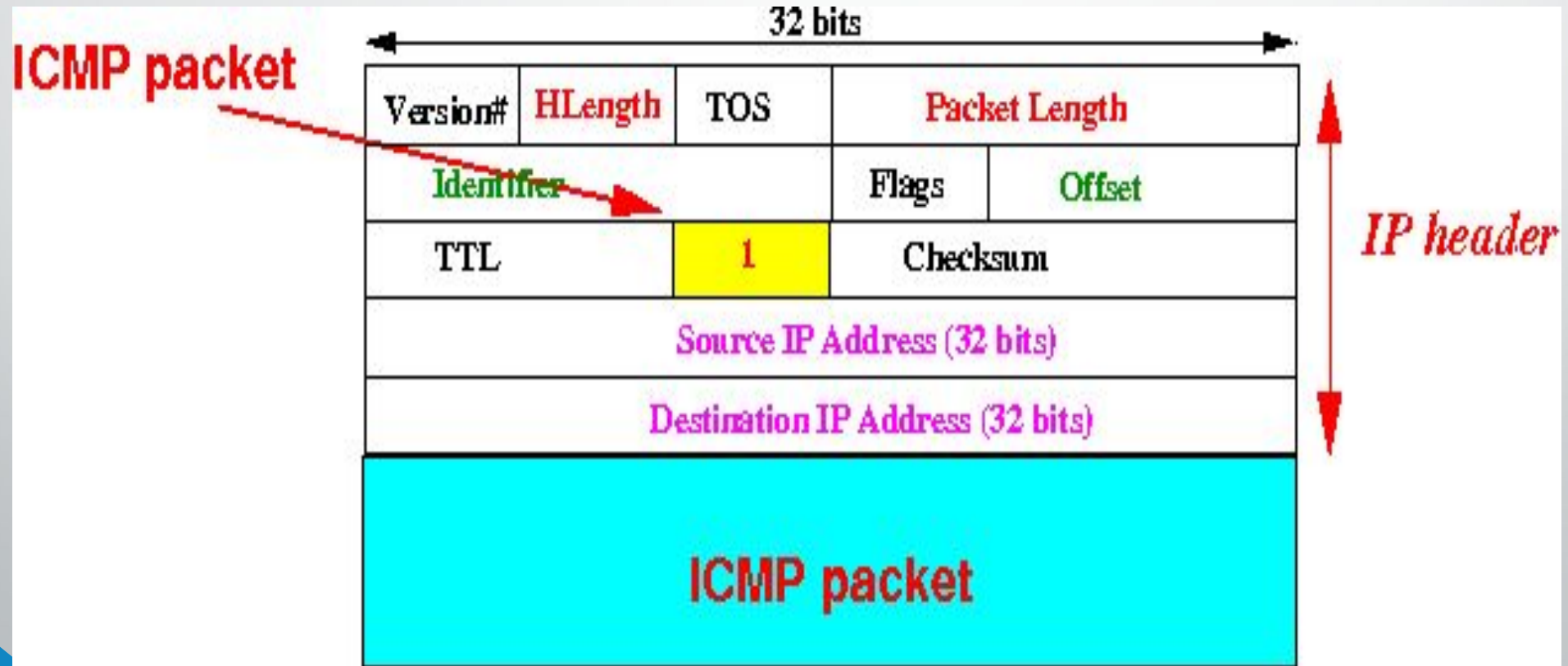
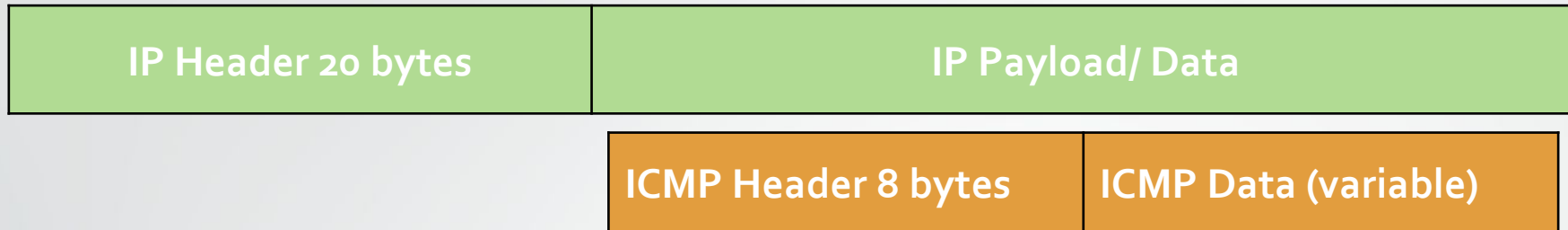
Pinging www.pepperdine.edu [137.159.8.186] with 32 bytes of data:
Reply from 137.159.8.186: bytes=32 time=37ms TTL=114
Reply from 137.159.8.186: bytes=32 time=38ms TTL=114
Reply from 137.159.8.186: bytes=32 time=36ms TTL=114
Reply from 137.159.8.186: bytes=32 time=38ms TTL=114

Ping statistics for 137.159.8.186:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 36ms, Maximum = 38ms, Average = 37ms

C:\USERS\LARRYP~1>
```

- Questions :
- Why 4 replies?
- What the time refer to?

ICMP Packet Format



ICMP Packet Format



ICMP Message Format

ICMP Type	ICMP Code	Description
0	0	Echo Reply (used by ping)
3	0	Destination Network Unreachable
3	1	Destination Host Unreachable
3	3	Destination Port Unreachable
8	0	Echo Request (used by ping)
11	0	TTL Expired (used by traceroute)

Type	Code	Description
0 – Echo Reply	0	Echo reply
3 – Destination Unreachable	0	Destination network unreachable
	1	Destination host unreachable
	2	Destination protocol unreachable
	3	Destination port unreachable
	4	Fragmentation needed and DF flag set
	5	Source route failed
5 – Redirect Message	0	Redirect datagram for the Network
	1	Redirect datagram for the host
	2	Redirect datagram for the Type of Service and Network
	3	Redirect datagram for the Service and Host
8 – Echo Request	0	Echo request
9 – Router Advertisement	0	Use to discover the addresses of operational routers
10 – Router Solicitation	0	
11 – Time Exceeded	0	Time to live exceeded in transit
	1	Fragment reassembly time exceeded
12 – Parameter Problem	0	Pointer indicates error
	1	Missing required option
	2	Bad length
13 – Timestamp	0	Used for time synchronization
14 – Timestamp Reply	0	Reply to Timestamp message

Unsuccessful Ping

```
C:\>ping 10.2.104.2
```

```
Pinging 10.2.104.2 with 32 bytes of data:
```

```
Request timed out.
```

```
Request timed out.
```

```
Request timed out.
```

```
Request timed out.
```

```
Ping statistics for 10.2.104.2:
```

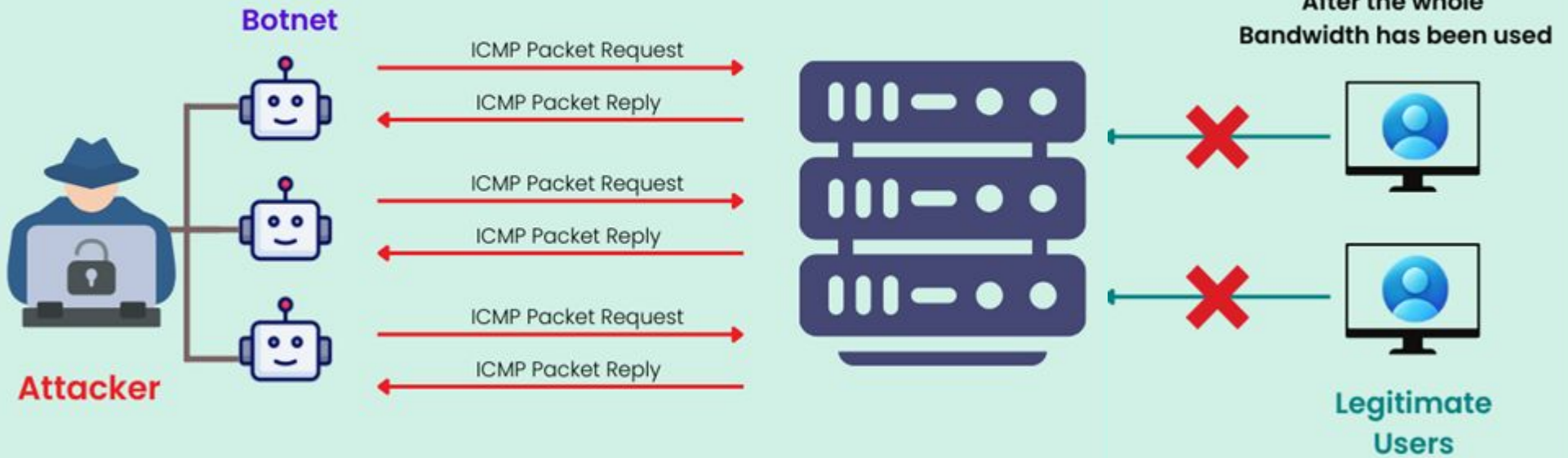
```
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Ping Attacks

- ICMP DDOS attack – Zombie Attack:

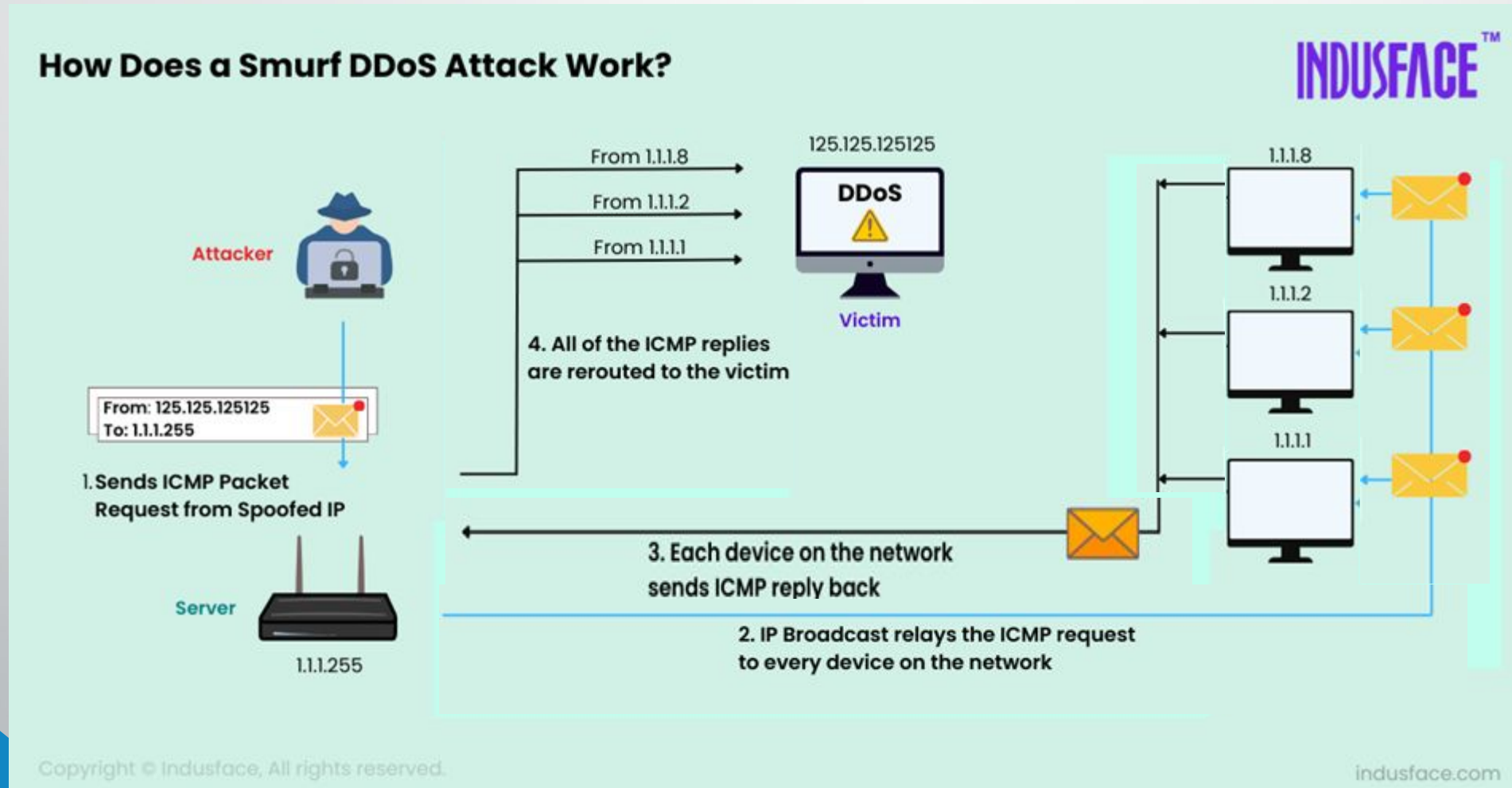
How Does a ICMP Flood DDoS Attack Work?

INDUSFACE™

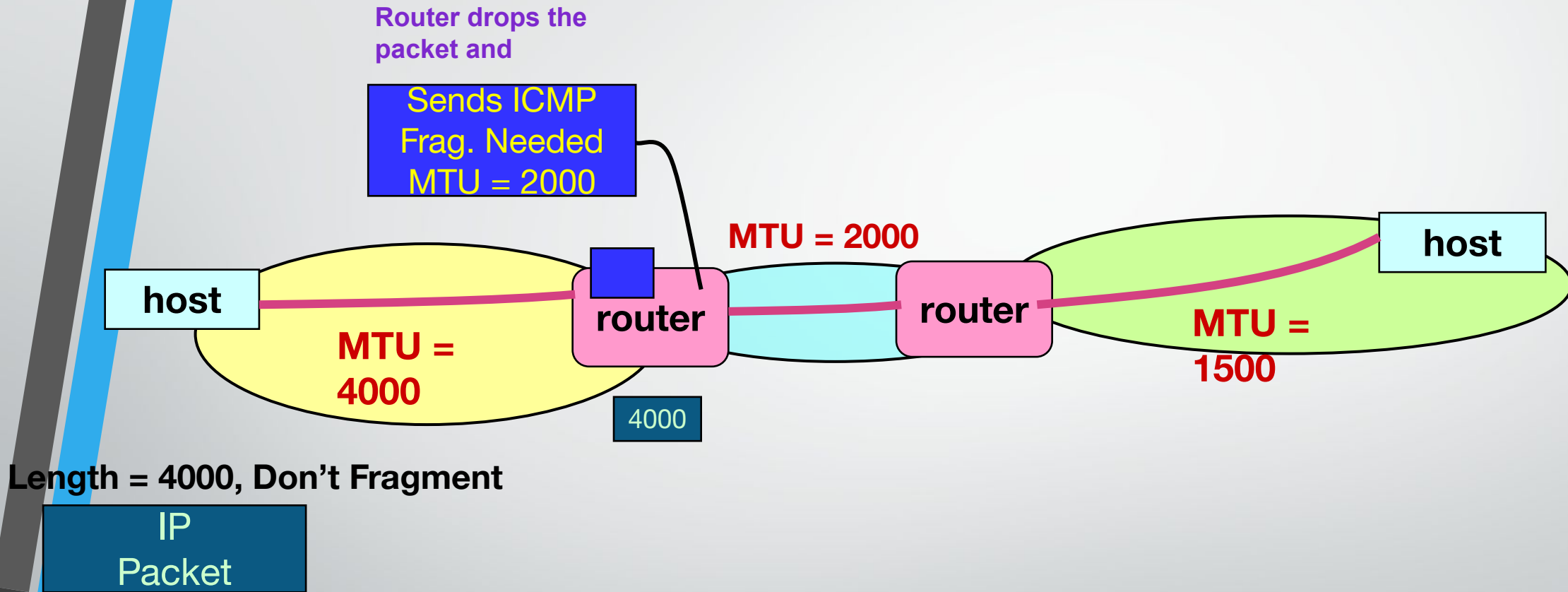


Ping Attacks

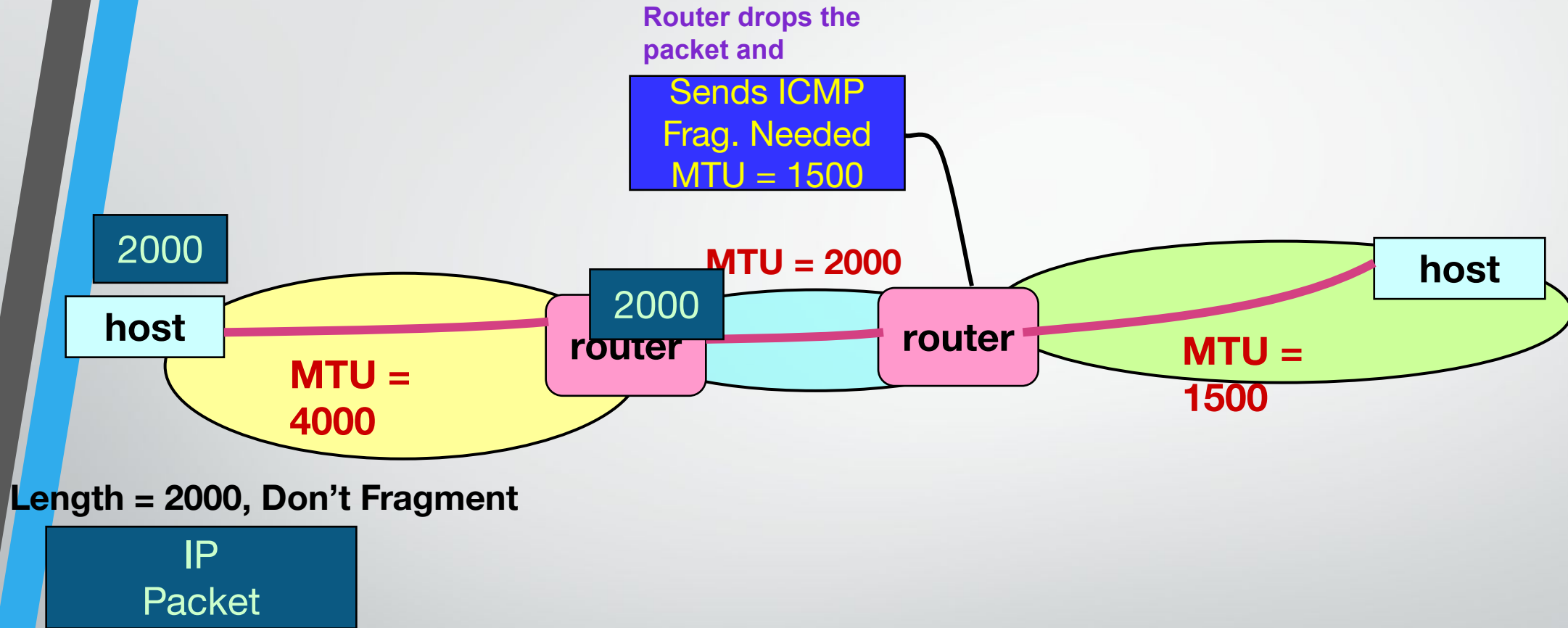
- ICMP DDOS attack – Packet magnification (or ICMP Smurf):



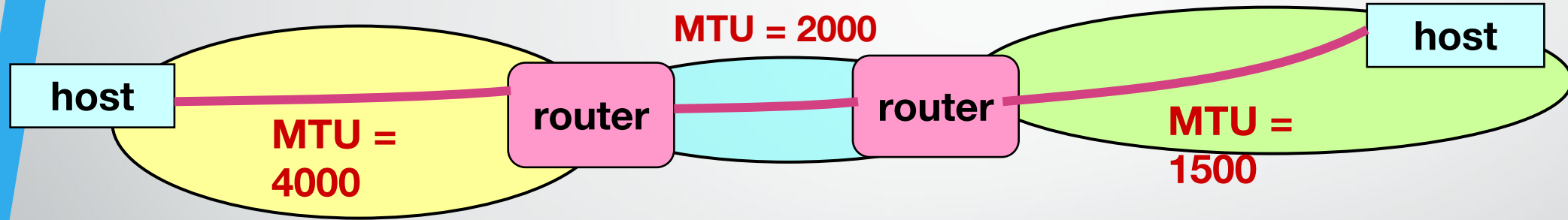
IP MTU Discovery with ICMP



IP MTU Discovery with ICMP



IP MTU Discovery with ICMP



Length = 1500, Don't Fragment

IP
Packet

When successful, no reply at IP level
"No news is good news"

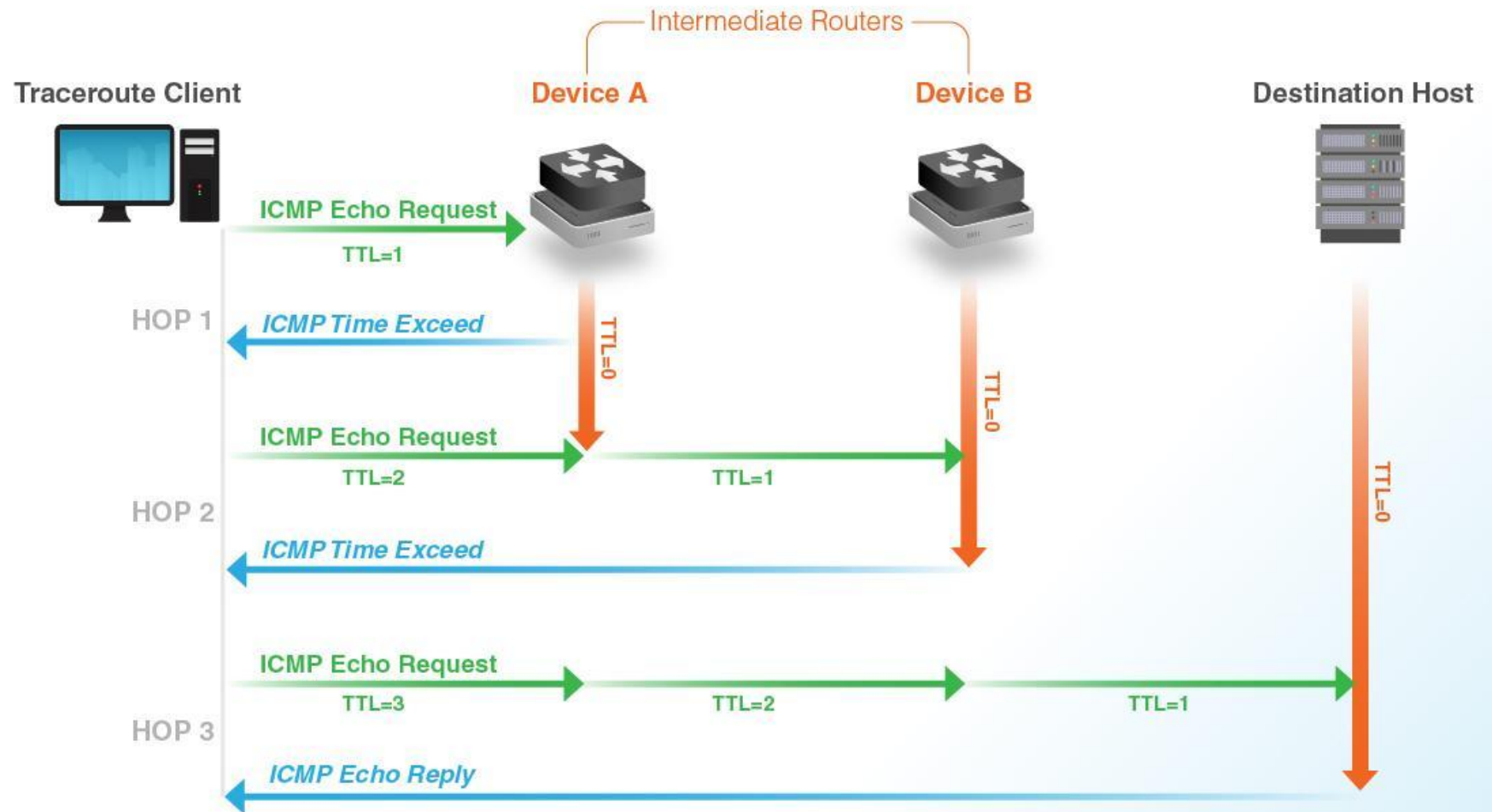
Higher level protocol might have some form of acknowledgement²⁶

Traceroute

- Shows the path your packets take through different routers to reach a destination.
- **Purpose:**
 - Identifies the routers or hops data passes through to reach its destination mostly for troubleshooting
- **Mechanism:**
 - Sends ICMP packets with TTL = 1, 2, 3...
 - Each router where TTL becomes 0 sends ICMP Time Exceeded
 - Traceroute uses these replies to list each hop in order



Traceroute



Traceroute

- **Results** The IP or hostname of every hop and the time each hop takes (latency)

```
Microsoft Windows [Version 10.0.22000.613]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>tracert www.google.com

Tracing route to forcesafesearch.google.com [216.239.38.120]
over a maximum of 30 hops:

  1  <1 ms    <1 ms    <1 ms    192.168.0.1
  2   2 ms     2 ms     2 ms
  3   5 ms     5 ms     7 ms    172.25.0.137
  4   5 ms     5 ms     5 ms    172.16.2.158
  5  18 ms    17 ms    17 ms    72.14.216.48
  6  18 ms    17 ms    17 ms    108.170.240.225
  7  17 ms    16 ms    17 ms    142.251.52.49
  8  18 ms    18 ms    18 ms    any-in-2678.1e100.net [216.239.38.120]

Trace complete.
```

- Commands:
 - Unix: **traceroute**
 - Cisco IOS: **traceroute (trace)**
 - DOS: **tracert**


```

Microsoft Windows [Version 10.0.19045.5131]
(c) Microsoft Corporation. All rights reserved.

C:\Users\skazi>tracert www.yahoo.com
'tracert' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\skazi>tracert www.yahoo.com

Tracing route to me-ycpi-cf-www.g06.yahoodns.net [27.123.42.205]
over a maximum of 30 hops:

  1      4 ms      1 ms      1 ms  172.18.192.1
  2      *        *        *      Request timed out.
  3      3 ms      1 ms      1 ms  172.31.2.129
  4      1 ms      1 ms      1 ms  10.151.6.89
  5      1 ms      2 ms      1 ms  10.0.100.5
  6      2 ms      1 ms      1 ms  202.4.100.253
  7      1 ms      2 ms      2 ms  GI0-2-2-aggr01.as58656.net [103.12.177.1]
  8      2 ms      2 ms      2 ms  10.12.176.237
  9      3 ms      2 ms      2 ms  103.16.155.149
 10      2 ms      1 ms      1 ms  103.16.152.30
 11     11 ms     11 ms     10 ms  103.16.152.82
 12      *        51 ms     51 ms  103.16.153.21
 13     51 ms     51 ms     51 ms  103.16.153.18
 14     57 ms     57 ms     57 ms  ae6-1538.rt.eqx.sin.sg.retn.net [87.245.240.208]
 15     62 ms     63 ms     62 ms  ix-be-20.ecore4.esin4-singapore.as6453.net [180.87.54.66]
 16     64 ms     65 ms     64 ms  if-bundle-18-2.qcore2.esin4-singapore.as6453.net [180.87.108.80]
 17     70 ms     70 ms     71 ms  180.87.55.59
 18      *        *        *      Request timed out.
 19     69 ms     70 ms     78 ms  14.143.59.46.static-mumbai.vsnl.net.in [14.143.59.46]
 20     68 ms     68 ms     67 ms  e2-ha.ycpi.ina.yahoo.com [27.123.42.205]

Trace complete.

C:\Users\skazi>_

```

Using Tracert

- Hop 1:** Our local router or gateway (private IP address).
- Hops 2–5:** Internal routing within Bracu ISP's private network (non-public IPs).
- Hop 6:** First public IP, ISP's gateway to the internet.
- Hops 7–9:** Routing through regional and backbone ISPs.
- Hops 10–13:** Routing through Singapore (a major internet hub).
- Hops 14–19:** Routing through Indian networks, ending in Mumbai.
- Hop 20:** Final destination—Yahoo's server, located in India, near Mumbai.

Traceroute: Another example

Hop 1: User LAN router

Hops 2-4: Verizon network (a backbone ISP)

Hops 5-6: Alternet (a backbone ISP)

Hops 7-11: Level 3 (a backbone ISP)

Hops 12-14: the Google LAN

```

C:\Windows\system32\COMMAND.com
C:\USERS\LARRYP~1>tracert www.google.com

Tracing route to www.l.google.com [74.125.19.147]
over a maximum of 30 hops:

 1      3 ms      1 ms      1 ms      192.168.1.1
 2     38 ms     37 ms     37 ms     L100.LSANCA-DSL-14.verizon-gni.net [71.105.96.11]
 3     38 ms     34 ms     36 ms     P1-3.LSANCA-LCR-03.verizon-gni.net [130.81.35.81]
 4     34 ms     37 ms     34 ms     so-6-1-2-0.LAX01-BB-RTR1.verizon-gni.net [130.81.28.225]
 5     37 ms     35 ms     38 ms     0.so-1-3-0.XL3.LAX15.ALTER.NET [152.63.114.145]
 6     36 ms     36 ms     40 ms     0.ge-6-0-0.BR2.LAX15.ALTER.NET [152.63.116.149]
 7     38 ms     40 ms     40 ms     xe-11-0-0.edge1.SanJose3.level3.net [4.68.111.249]
 8     46 ms     38 ms     49 ms     ae-73-70.ebr3.LosAngeles1.Level3.net [4.69.144.116]
 9     47 ms     55 ms     52 ms     ae-2.ebr3.SanJose1.Level3.net [4.69.132.9]
10     68 ms     54 ms    126 ms     ae-63-63.csw1.SanJose1.Level3.net [4.69.134.226]
11     72 ms     45 ms    115 ms     ae-1-69.edge1.SanJose1.Level3.net [4.68.18.14]
12    137 ms     51 ms     49 ms     GOOGLE-INC.edge1.SanJose1.Level3.net [4.79.43.146]
13     49 ms     49 ms     54 ms     209.85.251.98
14     47 ms     47 ms     46 ms     nuq04s01-in-f147.1e100.net [74.125.19.147]

Trace complete.

```

Traceroute: Request Timed Out

This message indicates that the router security settings keep it from revealing its identity or the router and connection are slow.

```
* * * Request timed out.
```

The End